

# Team [HY-Ru] – Technical Documentation

**29th RoboRacer Autonomous Racing Competition (IFAC 2026)** *Busan, Republic of Korea · August 24–27, 2026*

## 1. Team Information

**Team Name:** HY-Ru

**Affiliation:** Hanyang University, Republic of Korea

**Team Members:**

| Role        | Name           | Email                    |
|-------------|----------------|--------------------------|
| Team Leader | Seunghwan Shim | simseunghwan90@gmail.com |
| Member      | Chani Kim      | frogtoe03@gmail.com      |
| Member      | Hyunwoo Noh    | louisdarong2@gmail.com   |
| Member      | Youjin Lee     | eyoung050704@gmail.com   |
| Member      | Jeonghwan Choi | cdanni328@gmail.com      |
| Member      | Geunhyung Park | ghpark0905@gmail.com     |

## 2. Vehicle Hardware Stack

### 2.1 Chassis Platform

- **Model:** Traxxas Slash 4X4 (Gray Chassis, P/N 6822)
- **Scale:** 1:10, 4WD short-course truck
- **Dimensions (full vehicle, body + wheels):** Length 568 mm × Width 296 mm
- **Compliance:** Within competition limits (Width 238–341 mm, Length 454–654 mm) ✓
- **Rationale:** The Traxxas Slash 1:10 4WD platform is the de-facto standard chassis adopted by the F1TENTH/RoboRacer community, including the ICRA 2023 champion team ForzaETH. Its modular design, replaceable parts, and proven robustness make it ideal for autonomous racing development.

### 2.2 Powertrain

| Component                         | Specification                                                 |
|-----------------------------------|---------------------------------------------------------------|
| Drive Motor                       | Traxxas Velineon 3351R (brushless, single-unit configuration) |
| Steering Servo                    | DS3225 (25 kg·cm torque, metal gear)                          |
| Electronic Speed Controller (ESC) | Trampa VESC 6 MK VI                                           |
| Configuration                     | Single motor unit (compliant with regulations)                |

**Note on ESC choice:** The VESC 6 MK VI is selected for its open-source firmware, precise current/duty-cycle control, and seamless integration with ROS 2 via the `vesc_ros` driver. This enables low-level command interfaces (effort, velocity) required by high-performance controllers such as MAP.

## 2.3 Power System

- **Battery:** DXF 3S LiPo, 11.1 V, 5250 mAh, 140C (Model: M3s5250)
- **Configuration:** 1 × 3S LiPo (within the 4S maximum limit)
- **Rationale:** The 3S configuration provides a balance between sufficient peak current draw for aggressive maneuvers and weight optimization for the scaled platform.

## 2.4 Perception Sensors

- **LiDAR:** Hokuyo UST-10LX (single unit, compliant with one-LiDAR rule)
  - Scan range: up to 10 m
  - Scan angle: 270°
  - Angular resolution: 0.25°
  - Update rate: 40 Hz
- **Camera:** Not used (avoiding prohibited on-board detection / VIO output sensors)
- **IMU / Encoder:** [Integrated within VESC for odometry; additional IMU TBD]

The Hokuyo UST-10LX is the reference LiDAR used across the F1TENTH/RoboRacer community and provides reliable 2D scan data suitable for both SLAM-based localization and reactive obstacle detection.

## 2.5 Onboard Compute

- **Platform:** [NVIDIA Jetson Orin Nano]
- **OS:** Ubuntu 22.04 LTS
- **Middleware:** ROS 2 Humble Hawksbill

# 3. Software Stack

The software architecture follows the standard **Perception** → **Planning** → **Control** pipeline, with reference to the open-source [ForzaETH Race Stack](#) (JFR 2024).

### 3.1 Localization

- **Algorithm:** Particle Filter (Monte Carlo Localization)
- **Approach:** A pre-built occupancy grid map of the race track is generated offline via SLAM (e.g., SLAM Toolbox or Cartographer). During the race, the particle filter fuses LiDAR scan data and VESC-derived odometry to estimate the vehicle's pose in real time.
- **Rationale:** Particle filter localization is computationally lightweight, robust to dynamic environments (e.g., opponent vehicles, randomly placed obstacles), and is the localization method of choice for high-performance teams including ForzaETH.

### 3.2 Planning

The planning module is structured as a hierarchical pipeline:

#### (a) Global Planner — Raceline Optimization

- Offline computation of a minimum-curvature optimal raceline using the [TUM Global Racetrajectory Optimization](#) framework.
- Produces an optimal  $(x, y, \psi, \kappa, v)$  reference trajectory along with a forward-backward solved velocity profile respecting vehicle dynamic limits.

#### (b) Local Planner — Spliner

- The Spliner module from ForzaETH performs local trajectory generation in the **Frenet frame** relative to the global raceline.
- It generates collision-free spline-based deviations to the global trajectory when obstacles or opponents are detected ahead.
- Future extension: Integration of **Predictive Spliner** (RA-L 2024) for opponent trajectory prediction and proactive overtaking.

#### (c) Behavior Layer

- A state machine governs high-level decisions: TRACK\_GLOBAL → FOLLOW → OVERTAKE → RECOVERY.
- State transitions are triggered by perception inputs (obstacle proximity, opponent presence) and timing/safety conditions.

### 3.3 Control

- **Algorithm:** MAP (Model- and Acceleration-based Pursuit) Controller
- **Reference:** Becker et al., "*Model- and Acceleration-based Pursuit Controller for High-Performance Autonomous Racing*", ICRA 2023 (ForzaETH, ETH Zürich)
- **Description:** MAP extends classical Pure Pursuit by incorporating tire dynamics via a lateral-acceleration formulation derived from the L1 guidance law. Unlike geometric Pure Pursuit, which assumes negligible wheel slip, MAP uses a system-identified Pacejka tire model and converts guidance commands into steering angles via a lookup table.

- **Performance basis:** MAP achieves a four-fold reduction in lateral tracking error (0.055 m RMS at 11 m/s) compared to Pure Pursuit, and was a key contributor to ForzaETH's victory at the ICRA 2023 F1TENTH Grand Prix.
- **Rationale:** MAP offers near-MPC tracking performance while maintaining the computational simplicity required by the onboard embedded platform — a critical trade-off for scaled racing.

### 3.4 Detection and Avoidance

- **Detection Pipeline: LiDAR-based clustering + Kalman Filter tracking**
  - Incoming 2D LiDAR scans are pre-processed (range filtering, downsampling) and segmented via Euclidean clustering to extract candidate obstacle clusters.
  - Detected clusters are associated across frames and tracked using a Kalman Filter, providing state estimates ( $x$ ,  $y$ ,  $v_x$ ,  $v_y$ ) for each tracked obstacle.
- **Avoidance Pipeline: Spliner-based local replanning**
  - When a tracked obstacle lies within a safety corridor around the global raceline, the Spliner local planner generates a smooth spline deviation in the Frenet frame that avoids the obstacle while minimizing return time to the optimal raceline.
  - The avoidance maneuver respects vehicle dynamic constraints (curvature, velocity) inherited from the global trajectory optimizer.

## 4. System Architecture Summary

